



Dominant Psychosocial Response & Distress Classification in Oncology Messages

Interim - dataset, baselines, and plan

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PROJECT REVIEW

Where the project stands

- Motivation: triage the emotional state behind oncology messages - which response dominates, and how intense the distress.
- Task: 7-class psychosocial response + 3-level ordinal distress, predicted by separate models.
- Changed from the proposal: reframed “emotion” to “dominant psychosocial response”; added dual-LLM-judge auditing, quality tiers, and a strict label-leakage ban.
- Novelty: fine-grained response + ordinal distress on a fully synthetic, attribute-controlled corpus with auditing and blind human validation.

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PREVIOUS WORK

Four anchors from recent literature

Work	Focus	Relation to our project
Yu et al. 2023 (NeurIPS)	Attribute-conditioned synthetic generation; diversity vs. bias	Justifies our attribute-based generation (role, stage, type, tone, ...)
Li et al. 2023 (EMNLP)	Synthetic data is weaker for subjective tasks	Justifies our strict audit, quality tiers, and human check
Zheng et al. 2023 (NeurIPS)	LLM judges are scalable but biased (position, verbosity, ...)	Precedent for LLM evaluation; motivates two judges + human validation
Xu et al. 2026 (PLOS Dig. Health)	LLM annotation + emotional-tone classification of cancer posts	Closest domain baseline and the reference point for our novelty

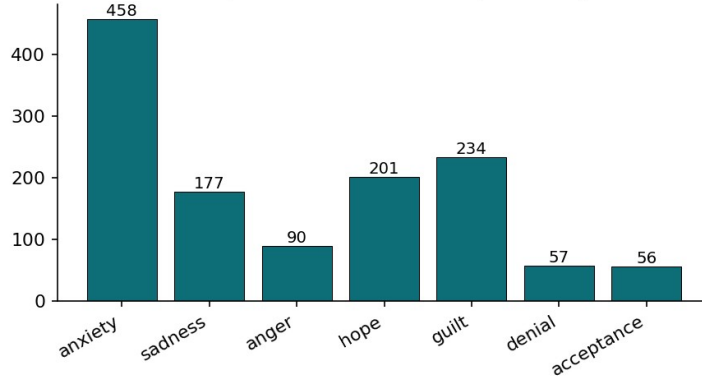
Novelty vs. Xu et al.: 7 response classes (not Neg/Neu/Pos) + a separate ordinal distress task + a synthetic attribute-controlled corpus + dual-judge tiers + human-ambiguity analysis.

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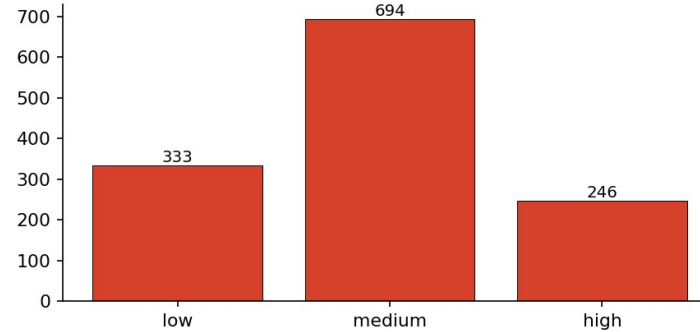
DATASET

A synthetic, audited corpus

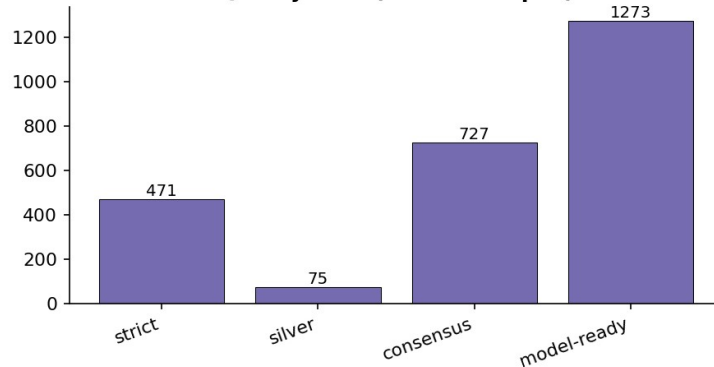
Response class balance (n=1273)



Distress level balance



Quality tiers (audited corpus)



Corpus at a glance

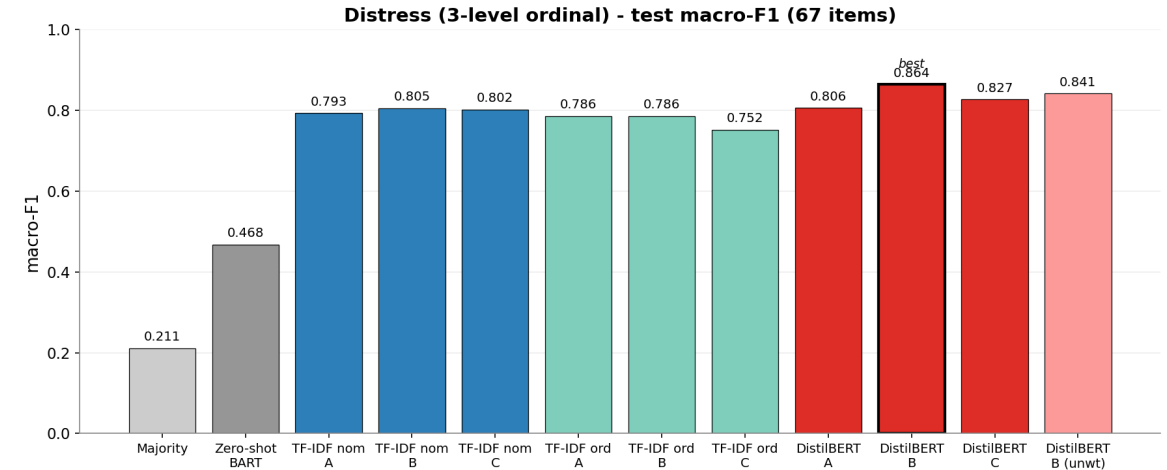
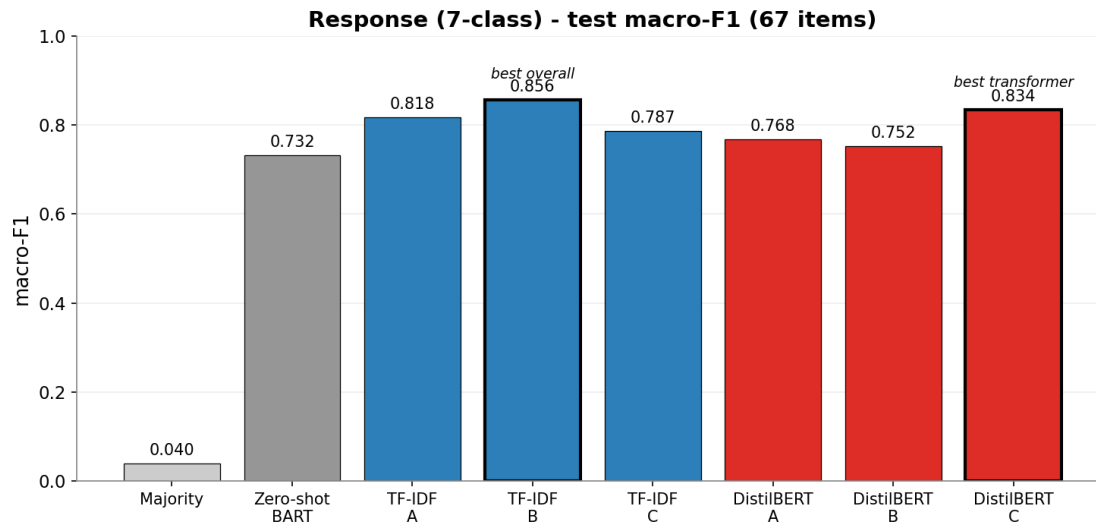
model-ready records: 1273
 message length: median 31 words (4-107)
 splits (seed 42): test 67 / val 45
 train_A 359 train_B 434 train_C 1161
 fully synthetic, label-leakage-controlled
 distributions reflect the generation pipeline,
 not real-world prevalence

- Corpus funnel: 2033 generated, 2025 valid audited, 1273 model-ready (used in experiments).
- Generated by attribute-based prompting; labeled by two blind LLM judges into strict / silver / consensus tiers.
- EDA reading: strong class imbalance (anxiety and medium-distress dominate); messages are short (median 31 words).
- Distributions reflect the generation pipeline, not real-world prevalence.

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BASELINE & RESULTS

Baselines and first results



- Off-the-shelf baseline = zero-shot BART. Both supervised families had higher point estimates than zero-shot; the gap was bootstrap-reliable for both families on distress and for TF-IDF-B on response.
- Error inspection: response shows an anger artifact (the model appears more closely aligned with the dual-judge policy than with the independent annotator's reading); distress error concentrates at the low / medium boundary.

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PLAN

Steps, scope and status

Phase	Scope	Status
Data generation	Attribute-based generation + rejection sampling	Done
Auditing & tiers	Dual-LLM-judge audit, strict / silver / consensus	Done
Splits & human check	Train/val/test + blind human annotation of test	Done
Modelling	TF-IDF, DistilBERT (A/B/C, 5 seeds), zero-shot	Done
Evaluation	Bootstrap CIs, error analysis, human reference	Done
Final presentation	Assemble repo, slides, visual abstract	In progress